Orbital Advection Across Azimuthal Level Jumps in the AthenaK Shearing Box

Implementation Document

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Abstract

With orbital advection (FARGO) on, AthenaK's shearing box could be refined only in complete azimuthal rings: every x_2 neighbor of a MeshBlock had to be at its own level. For clump-following adaptive refinement that is the dominant cost — a fragment in the $64H$ SC14 box refines a $64H$-long ring at every level. This document describes how the cell-centered orbital remap was extended so that an x_2 neighbor may be one level coarser or finer. Each block still remaps on its own grid; the neighbor's rows are restricted (from a finer neighbor) or prolongated (from a coarser one) to the block's resolution before the remap. The remap is rewritten in flux form, and an orbital-advection flux correction makes it exactly conservative across the level jump. A timestep limit keeps the shift within the immediate neighbor. The ring policy becomes an option. Uniform and ring-refined runs are bit-identical to the parent code. Hydro only: MHD with orbital advection on a refined mesh remains a fatal error. The tests are in the companion `fargo_amr_validation.pdf`.

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1 Scope and how to read this

This document describes mechanisms — data layout, index maps, call sites, guards — not results; those are in `fargo_amr_validation.pdf`. It builds on the shearing-box refinement work documented in `multigrid_sb_implementation.pdf` (Phase 0 there: the refinement policy, per-level orbital advection, the non-FARGO mode, AMR rebuilds of the shear state). That document now carries short pointers here where its statements about the ring policy are out of date.

1.1 Files touched

File	Change
`src/shearing_box/orbital_advection_cc.cpp`	Rewritten: pack (same, restricted, coarse-box messages), unpack with prolongation, remap fluxes G at the block faces, `SendFluxCC`, `RecvAndCorrectFluxCC`, `RowsAvailable`, `MaxTimeStep`
`src/shearing_box/orbital_advection.hpp`	`OrbitalAdvectionCC` buffers, sizes, MPI requests; virtual `InitRecv`/`ClearRecv`/`ClearSend`
`src/shearing_box/orbital_advection_tasks.cpp`	CC overrides of the three MPI functions (per-sub-face requests, flux receives)
`src/hydro/hydro{.hpp,_tasks.cpp}`	Tasks `SendFlx_OA`, `RecvFlx_OA`
`src/dust/dust{.hpp,_tasks.cpp}`	Same two tasks in the dust module's copy of the hydro task tail
`src/mesh/mesh.cpp`	Orbital-advection timestep limit in `Mesh::NewTimeStep`; policy warning text
`src/mesh/build_tree.cpp`	Ring check (I2) only with `oa_annular`
`src/mesh/mesh_refinement.{cpp,hpp}`	Ring-wise flag sync only with `oa_annular`
`src/hydro/hydro.cpp`, `src/mhd/mhd.cpp`	Comments
`src/pgen/tests/shwave.cpp`	Optional x_2 window for the moving-ring criterion (a moving box)

2 Background

2.1 Orbital advection in AthenaK

In the shearing box the background flow is $v_y = -q\Omega_0 x$. With orbital advection the state carries only the deviation from it, and the background transport is applied as a separate step: at the end of each time step (last RK stage, after the source terms) every cell is shifted in x_2 by

$$\Delta y = -q\Omega_0\, x_1\, \Delta t = (J + \epsilon)\, \Delta x_2, \qquad J = \mathrm{trunc}(\Delta y/\Delta x_2), \quad |\epsilon| < 1, \tag{1}$$

per x_2 column (fixed x_1, x_3). The integer part J is a copy, the fractional part a conservative remap with upwind "fluxes" F from a PLM or PPM reconstruction (`remap_fluxes.hpp`):

$$a_j^{\mathrm{new}} = a_{j-J} - \left(F_{j-J+1/2} - F_{j-J-1/2}\right). \tag{2}$$

`PackAndSendCC` sends the rows next to each x_2 face to the neighbor; `RecvAndUnpackCC` builds an extended column (own cells plus neighbor rows on both sides) in team scratch memory and applies Eq. (2). The tasks `SendU_OA`/`RecvU_OA` sit between the source terms and `RestrictU` of the last stage, so the shifted state then goes through the normal boundary exchange.

2.2 The annular policy and what it cost

The old pack used the same indices on both sides of a face ("neighbor must always be at same level") and read only neighbor slots 8 and 12, the first sub-face of each x_2 face. A coarser neighbor, which sits in sub-face slot 9–11 or 13–15, was not even read. Invariant I2 (complete x_2 rings at every level, with a ring-wise OR/AND of AMR flags) kept every x_2 neighbor at the same level. In the SC14 clump-following AMR runs ($64H \times 64H \times 16H$, 8^3 blocks) a fragment's refinement therefore covered the whole $64H$ azimuth at every level, reaching ∼28 000 blocks at $\beta = 4$ and 70–100 000 in the first $\beta = 3$ attempt.

2.3 Prior art: Athena++

Athena++'s orbital advection (T. Ono) supports refinement. Each block remaps on its own grid with an extended column; a finer neighbor sends restricted data, a coarser one a box of its own cells that the receiver prolongates with the standard limited-linear operator; a timestep limit keeps the shift within one neighbor block. Orbital advection runs as a separate stage with its own boundary cycle, so the prolongation uses fresh ghost cells. There is no flux correction: mass is conserved across level jumps only to truncation error. The design here follows the same pattern, adds the flux correction, and fuses the remap into the last RK stage as before.

3 Design

3.1 Three kinds of neighbor

2:1 balance guarantees that an x_2 face neighbor is at most one level away. Seen from a block on one of its x_2 faces (`OAFaceKind`):

Kind	Neighbor slots	What the block receives
same level	one, sub-face 0	the `nrow` rows next to the face, copied
finer	four (one per (x_1, x_3) quadrant)	from each fine block: its 2`nrow_fc` rows next to the face, restricted (2^3 averages) to `nrow_fc` coarse rows over its quadrant
coarser	one, sub-face $q = (l_{x1} \bmod 2) + 2(l_{x3} \bmod 2)$	a box of coarse cells over the block's quadrant, prolongated by the receiver

In every case the column handed to the remap is at the receiver's own resolution, so Eq. (2) and the reconstruction kernels are unchanged. The shift of a fine column uses the fine cell center, of a coarse column the coarse one; because Δy is linear in x_1, the two fine x_1 sub-columns of a coarse column move on average exactly as the coarse column does.

3.2 Buffers and messages

Buffers are flat arrays per (MeshBlock, face): `sbuf_oa`, `rbuf_oa` of shape $(n_{\mathrm{mb}}, 2,$ `sz_face`$)$. Face $f = 0$ is the inner x_2 face (neighbor slots 8–11), $f = 1$ the outer (12–15). The message for sub-face q (slot $8 + 4f + q$) starts at offset $q \cdot n_{\mathrm{var}} \cdot s$, where s is the per-variable size of that message kind; variable v starts at $v\,s$, and inside a variable the cells run with i fastest and rows in ascending j:

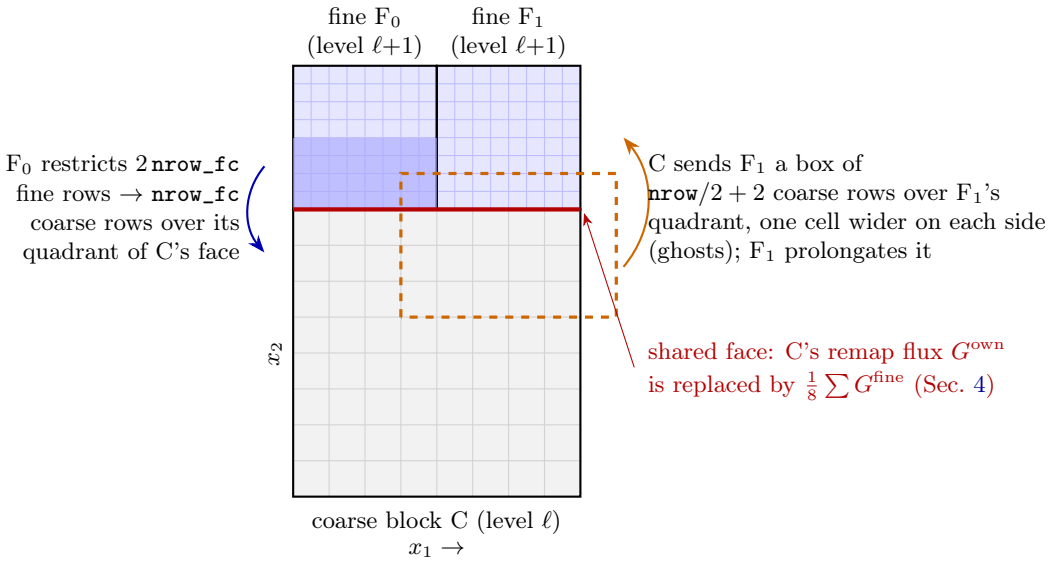


Figure 1: The two cross-level messages on an x_2 face, in the x_1–x_2 plane (the two further fine blocks of the face are behind, in x_3). Each block remaps on its own grid; the neighbor's rows arrive restricted or as a coarse box to prolongate. After the remap, the fine blocks send their face fluxes to C.

Message	Cells per variable	Rows (sender)
same level	$\texttt{sz_same} = n_{x3} \cdot \texttt{nrow} \cdot n_{x1}$	$j_s \dots j_s{+}\texttt{nrow}{-}1$ ($f{=}0$) or $j_e{-}\texttt{nrow}{+}1 \dots j_e$
fine → coarse	$\texttt{sz_fc} = \frac{n_{x3}}{2} \cdot \texttt{nrow_fc} \cdot \frac{n_{x1}}{2}$	$2\,\texttt{nrow_fc}$ fine rows next to the face, restricted
coarse → fine	$\texttt{sz_cf} = (\frac{n_{x3}}{2}{+}2) \cdot \texttt{nrow_cf} \cdot (\frac{n_{x1}}{2}{+}2)$	$j_s{-}1 \dots j_s{+}\texttt{nrow}/2$ or $j_e{-}\texttt{nrow}/2 \dots j_e{+}1$

with $\texttt{nrow_fc} = \min(\texttt{nrow}, n_{x2}/2)$ and $\texttt{nrow_cf} = \texttt{nrow}/2 + 2$, and

$$\texttt{sz_face} = n_{\text{var}} \max\left(\texttt{sz_same},\ 4\max(\texttt{sz_fc}, \texttt{sz_cf})\right).$$

The coarse box of quadrant (f_x, f_z) spans $i = i_s + f_x n_{x1}/2 - 1 \dots i_s + (f_x{+}1)n_{x1}/2$ and the same in k. Its outermost cells (the row across the face and the transverse layer at the block edge) are ghost cells.

A message lands in the receiver's buffer at the receiver's sub-face offset, read from `nghbr.dest`. Senders on the same rank write straight into the receiver's buffer; otherwise the message goes through `sbuf_oa` and `MPI_Isend` with tag $(\texttt{dest} \ll \texttt{NUM_BITS_LID}) \,|\, \texttt{lid}$ on `comm_orb_advect`, as the other boundary exchanges do. Receives are posted per sub-face in `OrbitalAdvectionCC::InitRecv`, with the size of the kind seen from the receiver; MPI requests are kept per (block, sub-face), $8n_{\text{mb}}$ of each.

3.3 Unpacking: index maps

The extended column has $n_{x2} + 2\,\texttt{nrow}$ cells; the block's own cells start at $j_f = \texttt{nrow}$. The `nrow` rows beyond face f are numbered $t = 0 \dots \texttt{nrow} - 1$ in ascending j (so on the inner face $t = \texttt{nrow} - 1$ touches the block, on the outer face $t = 0$ does). Row t is loaded by:

- **same level:** the received row t.
- **finer:** quadrant $f_x = \lfloor (i - i_s)/(n_{x1}/2) \rfloor$, f_z likewise; coarse row $t_c = t - (\texttt{nrow} - \texttt{nrow_fc})$ on the inner face, $t_c = t$ on the outer. Rows with t_c outside $[0, \texttt{nrow_fc})$ are never read (guarded, Sec. 5).

- **coarser:** with the distance d from the face ($d = \texttt{nrow} - 1 - t$ on the inner face, $d = t$ on the outer), the coarse cell is at box row $jj = \texttt{nrow}/2 - \lfloor d/2 \rfloor$ (inner) or $\lfloor d/2 \rfloor + 1$ (outer), box column $ii = \lfloor (i - i_s)/2 \rfloor + 1$, box layer kk likewise. The value is

$$c + s_1\,\delta_1 + s_2\,\delta_2 + s_3\,\delta_3, \qquad \delta_n = \tfrac{1}{8}\left[\operatorname{sgn}(\Delta_n^-) + \operatorname{sgn}(\Delta_n^+)\right] \min(|\Delta_n^-|, |\Delta_n^+|), \tag{3}$$

  the `ProlongCC` formula ($\Delta_n^\pm$ the one-sided differences of the coarse box in direction n), with $s_1 = -1$ for even $i - i_s$ and $+1$ for odd, s_3 likewise in k, and $s_2 = +1$ for the fine row in the upper half of the coarse cell (d even on the inner face, odd on the outer).

The ghost cells used for the slopes hold the state of the previous stage's exchange: for RK2 they differ from the updated state by $O(\Delta t^2)$, and they only enter minmod-limited slopes. The remap shifts the coarse-box values like any other, and their mean over each coarse cell is the coarse value, so the prolongation conserves.

4 The orbital-advection flux correction

4.1 The remap in flux form

Eq. (2) telescopes: with

$$G_{j-1/2} = \begin{cases} F_{j-J-1/2} + \displaystyle\sum_{l=j-J}^{j-1} a_l, & J \geq 0, \\ F_{j-J-1/2} - \displaystyle\sum_{l=j}^{j-J-1} a_l, & J < 0, \end{cases} \qquad a_j^{\text{new}} = a_j - \left(G_{j+1/2} - G_{j-1/2}\right), \tag{4}$$

$G_{j-1/2}$ is the total amount (per cell volume) moved in $+x_2$ through face $j - 1/2$: the $|J|$ whole cells it passes plus the fractional flux. In the unpack kernel, after the fractional fluxes are known, one thread per column evaluates G at the block's two x_2 faces and stores it in $\texttt{gflx}(m, f, v, k, i)$ (multilevel meshes only).

4.2 Why the two sides disagree, and the correction

Across a level jump the coarse block remaps restricted fine data with its column velocity, and each fine block remaps prolongated coarse data with its own. Both approximate the same transport, but not identically, so without correction the mass that leaves one side is not the mass that enters the other (the mass drift is 10^{-7}–10^{-6} in the tests). The fine side is taken as the better estimate, as in the hydro flux correction. A coarse column covers 2×2 fine columns and the volume ratio is $1/8$, so the amount through the coarse face is

$$G^{\text{fine}} = \tfrac{1}{8} \sum_{2\times 2} G_f. \tag{5}$$

`SendFluxCC` (task `SendFlx_OA`, after `RecvU_OA`) computes this on every block whose x_2 neighbor is coarser and sends it ($\frac{n_{x3}}{2} \cdot \frac{n_{x1}}{2}$ values per variable; separate communicator `comm_orb_flx`). `RecvAndCorrectFluxCC` (task `RecvFlx_OA`) replaces the coarse block's own flux in the row next to the face:

$$a_{j_s} \mathrel{+}= G^{\text{fine}} - G^{\text{own}} \quad \text{(inner face)}, \qquad a_{j_e} \mathrel{+}= G^{\text{own}} - G^{\text{fine}} \quad \text{(outer face)}. \tag{6}$$

The two faces are separate kernels. The receives are posted with the data receives in `InitRecv`, waited for in `ClearRecv`. `<shearing_box> oa_flux_correct = false` switches the correction off (for tests). The dust module builds its own copy of the hydro task tail (`dust_tasks.cpp`); the two tasks are inserted there too, or the posted flux receives would never be matched.

5 How far the shift may reach

5.1 Rows available, and the guard

The PPM reconstruction of the upwind cell reads two cells on either side of it, so the remap of a block with integer shift J reads $|J|+3$ rows from its upstream neighbor (and $\max(0, 3-|J|)$ from the other). All of them must come from the immediate x_2 neighbor. `RowsAvailable`(m) returns

$$n_{\mathrm{avail}} = \begin{cases} \texttt{nrow} & \text{neither } x_2 \text{ neighbor finer,} \\ \min(\texttt{nrow}, n_{x2}/2) & \text{a finer } x_2 \text{ neighbor (a fine block is } n_{x2}/2 \text{ coarse rows).} \end{cases} \tag{7}$$

`nrow` defaults to $\min(n_{x2}, \texttt{maxjshift} + 3)$ (rounded up to even on multilevel meshes, so that it is a whole number of coarse rows) and can be set with `<shearing_box> oa_nrow` ($\le n_{x2}$). `RecvAndUnpackCC` checks every step, with this step's Δt and each block's $\max|x_1|$, that $\lfloor q\Omega_0 \Delta t \max|x_1|/\Delta x_2 \rfloor + 3 \le n_{\mathrm{avail}}$, and aborts naming the block otherwise.

5.2 The timestep limit

The guard alone would end a long AMR run the first time the mesh put a fine block next to a coarse one at large $|x_1|$. `OrbitalAdvectionCC::MaxTimeStep` therefore returns

$$\Delta t_{\mathrm{OA}} = \min_m \; 0.999 \, \frac{(n_{\mathrm{avail},m} - 2)\, \Delta x_{2,m}}{q\Omega_0 \max|x_1|_m}, \tag{8}$$

which `Mesh::NewTimeStep` takes into the cycle timestep (not multiplied by the CFL number), for hydro and MHD alike. The driver calls `Mesh::NewTimeStep` after the AMR update, so the limit is evaluated on the mesh the next step will use. It binds only when shifts are large. On 8^3 blocks a coarse block next to a finer x_2 neighbor allows one coarse cell per step: at $|x| = 32H$ and 4 cells per H on the coarse side, $\Delta t \le 0.0104/\Omega_0$ at the root level and $2^{-\ell}$ times that at level ℓ. In the SC14 runs the hydro timestep is well below this at the root level; on refined levels the two can be comparable.

5.3 The old buffer depth

The previous code allocated and packed `ng` + `maxjshift` rows per face, with `maxjshift` $= \lfloor \mathrm{CFL}\, q\Omega_0 \max|x_1| \rfloor + 1$, doubled on multilevel meshes, and checked the shift against `maxjshift`. For the 64H SC14 box on 8^3 blocks that is $4 + 30 = 34$ rows, so the pack read far outside the 8-row block (past the end of the array for the last block). The results were not affected, because the actual shift stayed below one cell and those rows were never used. Buffers are now at most one block deep (`nrow` $\le n_{x2}$), about $4\times$ smaller for 8^3 blocks, and the guard compares the shift with what the neighbor can actually supply.

6 Policy and control flow

6.1 The refinement policy

- **I1** (one uniform level along both shear-periodic x_1 faces) is unchanged.
- **I2** (complete x_2 rings) is checked in `Mesh::CheckShearingBoxRefinement` only with `<shearing_box> oa_annular = true`; the default is `false`.
- The **ring-wise flag synchronization** in `MeshRefinement::CheckForRefinement` runs only with `oa_annular = true`. User criteria such as the `gravito_turb` hold/buffer criterion then refine what they flag.
- The **graded cap** and the x_1-boundary veto are unchanged.

6.2 Task sequence of the last RK stage

Task	With orbital advection
`RKUpdate`, `HydroSrcTerms`	as before
`SendU_OA`	`PackAndSendCC`: same / restricted / coarse-box messages
`RecvU_OA`	`RecvAndUnpackCC`: wait, guard, load + prolongate, remap, store G
`SendFlx_OA`	`SendFluxCC`: restricted G to coarser x_2 neighbors
`RecvFlx_OA`	`RecvAndCorrectFluxCC`: replace the face flux on coarse blocks
`RestrictU`, `SendU`, ...	as before

The MHD task list calls the same `OrbitalAdvectionCC` for the cell-centered variables but not the two flux tasks: with MHD, orbital advection on a refined mesh is fatal, so no level jump can occur.

6.3 A test criterion: the moving box

The shwave problem generator's prescribed moving-ring criterion (`MovingRingRefine`) takes an optional x_2 window, $y_c(t) = y_{c0} + A_y \sin(\omega_y t) + v_y t$, wrapped periodically into $[x_{2,\min}, x_{2,\max})$. With `ring_yhwidth` > 0 a block is refined only if it overlaps both the x_1 interval and the x_2 window (or one of its periodic images): a box that moves in x_1 and drifts through x_2.

7 Input-file interface

7.1 Parameters

Block	Parameter	Default	Meaning
`<shearing_box>`	`oa_annular`	`false`	`true`: complete x_2 rings required (I2) and AMR flags synchronized ring-wise
	`oa_flux_correct`	`true`	Orbital-advection flux correction at x_2 level jumps (`false` for tests only)
	`oa_nrow`	$\min(n_{x2}$, `maxjshift` $+3)$	Rows sent across a same-level x_2 face; at most n_{x2}
`<problem>` (shwave)	`ring_yc0`, `ring_yamp`	0	Window center and oscillation amplitude
	`ring_yfreq`, `ring_yvel`	1, 0	Oscillation frequency and drift speed
	`ring_yhwidth`	-1	> 0: refine only inside the x_2 window

Command-line overrides replace only parameters already in the input file, so `oa_annular` and `oa_flux_correct` must be written into a copy of the input when they are to be changed and are not there.

7.2 Reference inputs

File (under `inputs/`)	Configuration
`shearing_box/epicycle_smr_ypatch.athinput`	Epicycle, level-1 patch over part of x_2
`shearing_box/hydro_incompress_shwave_smr_ypatch.athinput`	Vortical shwave, static patch
`shearing_box/hydro_incompress_shwave_amr_box.athinput`	Vortical shwave, moving box (AMR)
`tests/swing_selfgrav_smr_ypatch.athinput`	Swing with multigrid gravity, patch

8 Guard reference

Location	Condition	Protects
`OrbitalAdvectionCC::RecvAndUnpackCC`	integer shift $+3 >$ rows available from the x_2 neighbors	The remap stencil stays inside the immediate neighbor; normally prevented by the timestep limit
`Mesh::CheckShearingBoxRefinement`	incomplete x_2 ring with `oa_annular = true`	The optional annular policy
`Mesh::CheckShearingBoxRefinement`	x_1-face blocks on more than one level	I1 (unchanged)
`MHD::MHD`	orbital advection + refinement	The face-field remap at level jumps (unchanged)

9 Limitations and next steps

1. **MHD.** The face-centered field remap (`orbital_advection_fc.cpp`, EMF form) needs the same cross-level exchange plus a divergence-free treatment at the level jump (prolongation of face

fields, EMF correction of the orbital EMFs). Until then MHD + orbital advection + refinement stays fatal.

2. **One neighbor deep.** All upstream data come from the immediate x_2 neighbor; the timestep limit enforces it. Reaching through two neighbors would lift the limit on small blocks far from $x = 0$.

3. **Previous-stage ghosts in the coarse box.** They enter only the limited slopes of the prolongation; a separate orbital-advection stage with its own boundary cycle (as in Athena++) would remove this at the cost of an extra exchange.

4. **Non-FARGO refinement with x_2 level jumps** collapses at the first step ($\Delta t \to 10^{-45}$) — in the parent code too, with or without `prolong_primitives`. It is outside this work.

5. **Production test pending.** The SC14 clump-following AMR with and without rings (block count and cost) has not been run yet.